Brice MARTIN

FINAL YEAR PHD STUDENT AT ISAE - SUPAERO

Seeking a position in the fields of CFD and Machine Learning

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FORMATION

Artificial intelligence applied to fluid dynamics, PhD ISAE-SUPAERO

2020 - 2024 Toulouse

- Aerodynamics, CFD, Gust mitigation
- Machine learning: Deep Reinforcement Learning,CNN

Fluid dynamics and energetics, MSc ENSEEIHT

2017 - 2020 Toulouse

- Turbulence, Aerodynamics, Multi-phase flows, CFD
- Machine learning, Neural network, Data assimilation

EXPERIENCES

PhD — ISAE-SUPAERO

October 2020 - ongoing Toulouse

- Built gust mitigation technique based on deep reinforcement learning
- Performed data assimilation for gust estimation

Internship — Onera

April - August 2020 Toulouse

- Prevented the ice formation on aircraft wings
- Carried out numerical study on lubrication phenomena

Project — Tekam ingénierie and SNCF

September - march 2020 Toulouse

- Performed flow simulations on STARCCM
- Optimized the train drag coefficient

Internship — Institute of chemical process fundamental

July - August 2019 Prague

- Studied vortex-bubble interaction
- Conducted experimental and numerical flow analysis

SKILLS

Aerodynamic | CFD | Machine learning | Deep Reinforcement Learning

SOFTWARE

Linux | Star-CCM | Python | Fluent

STUDENT PROJECT

- Image Classification using Deep Neural Networks with Keras
- Development of CFD solvers
- Machine learning for flood forecasting

LANGUAGES

French
English
Spanish

PUBLICATION

- *Empirical evaluation of robustness of DRL algorithms for a pitch control problem*, Brice Martin, Thierry Jardin, Emmanuel Rachelson, Michael Bauerheim. Presented at the 3AF 56th conference.
- *Vortex gust mitigation using onboard measurements using deep reinforcement learning*, Brice Martin, Thierry Jardin, Emmanuel Rachelson, Michael Bauerheim. Under review at Data-Centric Engineering

HOBBIES

- Theatre, improvisational theatre
- Sports, cycling, skiing